AUGMENTED DECISION

Statistical learning and classification

Empirical risk, generalization, and minimax limits

Lecture 2 / Three hours, including a 15-minute break

A theoretical extension of the integrated syllabus. Definitions, theorem assumptions, and proof sketches.

Training, validation, and test samples

Training	Fit the parameters or construct candidate predictors.
Validation	Select a model class, tuning parameter, or decision threshold.
Test	Evaluate the final frozen procedure on unused observations.

All parts represent the same target population in today's theory. They contain independent observations.

A split separates statistical roles. It does not repair distribution shift or dependence.

Why independence matters

Condition on every choice made using training and validation data.

- The final predictor is then fixed relative to a fresh test sample.
- Its test loss can estimate its population loss.
- Selecting the best test score makes the reported score part of model selection.
- Refitting after validation is allowed if a final untouched test sample remains.

The information used by the full procedure determines whether evaluation is independent.

Underfitting and overfitting

Underfitting: the candidate class cannot represent a sufficiently good rule, or the fitting algorithm fails to reach one.

Overfitting: adaptation to the observed sample produces an overly optimistic fit and poor performance on new cases.

- A constant classifier may miss a simple decision boundary.
- A lookup table can memorize random training labels.
- Larger classes reduce approximation error, but need not reduce estimation error.

These are diagnoses. Training and validation curves are evidence, not mathematical definitions or guaranteed shapes.

The statistical learning experiment

$$Z = (X, Y) \sim P, \qquad S_n = (Z_1, \ldots, Z_n) \sim P^n.$$

- P is the unknown distribution on $\mathcal{X} \times \mathcal{Y}$.
- $\mathcal{F}$ is a fixed class of measurable predictors $f : \mathcal{X} \to \mathcal{A}$.
- $\ell(a, y)$ is the loss of action a when the response is y.
- A learning algorithm maps S_n to $\hat{f} \in \mathcal{F}$.

n: training size. m: test size. $\alpha \in (0, 1)$: failure probability. p: feature dimension. v: VC dimension. All logarithms are natural.

Population risk, gap, and excess risk

$$
\begin{aligned}
R_P(f) &= \mathbb{E}_{Z\sim P}\ell(f(X), Y),\\
\widehat{R}_n(f) &= \frac{1}{n}\sum_{i=1}^{n} \ell(f(X_i), Y_i),\\
G_n(f) &= R_P(f) - \widehat{R}_n(f).
\end{aligned}
$$

The **generalization error** means population risk $R_P(f)$ here.
The **generalization gap** is $G_n(f)$ and may have either sign.

$$
R_P^* = \inf_{f \text{ measurable}} R_P(f), \qquad \mathcal{E}_P(f) = R_P(f) - R_P^*.
$$

Empirical risk minimization

$$\hat{f} \in \arg\min_{f \in \mathcal{F}} \widehat{R}_n(f).$$

An approximate ERM satisfies, for $\varepsilon_{\mathrm{opt}} \geq 0$,

$$\widehat{R}_n(\hat{f}) \leq \inf_{f \in \mathcal{F}} \widehat{R}_n(f) + \varepsilon_{\mathrm{opt}}.$$

- The hypothesis class restricts the actions available to the learner.
- The empirical objective approximates an unknown population objective.
- Optimization accuracy alone does not establish statistical accuracy.

Approximation and estimation error

Let $R^*_{P,\mathcal{F}} = \inf_{f \in \mathcal{F}} R_P(f)$. Then

$$\mathcal{E}_P(\widehat{f}) = \underbrace{R^*_{P,\mathcal{F}} - R^*_P}_{\text{approximation error}} + \underbrace{R_P(\widehat{f}) - R^*_{P,\mathcal{F}}}_{\text{estimation / fitting error}} .$$

For nested classes $\mathcal{F}_1 \subseteq \mathcal{F}_2$, approximation error cannot increase.

No corresponding monotonicity holds for the fitted predictor's population risk.

The statistical problem is to balance representational error and learning error.

A deterministic ERM oracle inequality

Define $\Delta_n(\mathcal{F}) = \sup_{f \in \mathcal{F}} |R_P(f) - \widehat{R}_n(f)|$. For a population minimizer $f^*_{\mathcal{F}}$,

$$\begin{aligned} R_P(\hat{f}) - R_P(f^*_{\mathcal{F}}) &= [R_P(\hat{f}) - \widehat{R}_n(\hat{f})] \\ &\quad + [\widehat{R}_n(\hat{f}) - \widehat{R}_n(f^*_{\mathcal{F}})] \\ &\quad + [\widehat{R}_n(f^*_{\mathcal{F}}) - R_P(f^*_{\mathcal{F}})] \\ &\leq 2\Delta_n(\mathcal{F}) + \varepsilon_{\mathrm{opt}}. \end{aligned}$$

Uniform control of the empirical process gives an excess-risk bound.

An approximating sequence gives the same conclusion if the infimum is not attained.

Hoeffding concentration for a fixed predictor

Assume $0 \le \ell(f(X), Y) \le 1$ and f is fixed independently of S_n.

$$\mathbb{P}\Big(|\widehat{R}_n(f) - R_P(f)| > t\Big) \le 2e^{-2nt^2}.$$

Thus, with probability at least $1 - \alpha$,

$$|\widehat{R}_n(f) - R_P(f)| \le \sqrt{\frac{\log(2/\alpha)}{2n}}.$$

Substituting a training-dependent $\hat{f}$ into a fixed-f theorem is unjustified.

For loss in $[0, B]$, multiply the bound by B. Gaussian squared loss is unbounded.

Uniform convergence for a finite class

Assume $|\mathcal{F}| = M < \infty$, fixed before observing S_n, and $0 \le \ell \le 1$.

$$\begin{aligned}\mathbb{P}\Big(\Delta_n(\mathcal{F}) > t\Big) &\le \sum_{f \in \mathcal{F}} \mathbb{P}\Big(|R_P(f) - \widehat{R}_n(f)| > t\Big) \\ &\le 2Me^{-2nt^2}.\end{aligned}$$

Consequently, with probability at least $1 - \alpha$,

$$\Delta_n(\mathcal{F}) \le \sqrt{\frac{\log(2M/\alpha)}{2n}}.$$

A union bound requires no independence across the predictors' losses.

Finite-class ERM generalization bound

Under the preceding assumptions, with probability at least $1 - \alpha$,

$$R_P(\hat{f}) - R^*_{P,\mathcal{F}} \leq \sqrt{\frac{2\log(2M/\alpha)}{n}} + \varepsilon_{\text{opt}}.$$

Adding approximation error gives a bound relative to the Bayes risk:

$$\mathcal{E}_P(\hat{f}) \leq (R^*_{P,\mathcal{F}} - R^*_P) + \sqrt{\frac{2\log(2M/\alpha)}{n}} + \varepsilon_{\text{opt}}.$$

Bounds larger than the loss range are valid but uninformative. [SL, Chapter 4]

Proof exercise: sample size for finite-class ERM

Let $M = 1000$, $n = 1000$, $\alpha = 0.05$, and $\varepsilon_{\text{opt}} = 0$.

1. Calculate the uniform deviation bound and the excess-risk bound.
2. Solve for a sufficient n to ensure excess risk at most 0.10 with probability at least 0.95.
3. Explain why this does not guarantee population risk at most 0.10.

Work under the finite fixed class, i.i.d. sample, and bounded-loss assumptions.

Exercise solution: the comparator matters

$$\sqrt{\frac{\log(40000)}{2000}} \approx 0.0728, \qquad 2\Delta_n \lesssim 0.1456.$$

For target excess risk $\varepsilon_{\mathrm{tar}} = 0.10$, a sufficient sample size is

$$n \geq \frac{2\log(2M/\alpha)}{\varepsilon_{\mathrm{tar}}^2} = 2119.33\ldots, \quad \text{hence } n = 2120.$$

The conclusion is $R_P(\widehat{f}) \leq R^*_{P,\mathcal{F}} + 0.10$.

The best available rule may already have substantial population risk.

Independent test error: a conditional theorem

Let $T_m \sim P^m$ be independent of all fitting and selection data D. The final $\hat{f} = A(D)$ is fixed conditional on D.

$$\widehat{R}_{T_m}(\hat{f}) = \frac{1}{m} \sum_{j=1}^{m} \ell(\hat{f}(X'_j), Y'_j).$$

If $0 \leq \ell \leq 1$, then conditional on D,

$$\mathbb{E}[\widehat{R}_{T_m}(\hat{f}) \mid D] = R_P(\hat{f}),$$

$$\mathbb{P}\left(\left| \widehat{R}_{T_m}(\hat{f}) - R_P(\hat{f}) \right| > \sqrt{\frac{\log(2/\alpha)}{2m}} \,\middle|\, D \right) \leq \alpha.$$

Validation among a fixed set of candidates

Train K candidates $f_1, \ldots, f_K$ without using an independent validation sample of size n_{val}.

$$\hat{k} \in \arg\min_{k \le K} \widehat{R}_{\mathrm{val}}(f_k).$$

For $0 \le \ell \le 1$, conditionally on training data, with probability at least $1 - \alpha$,

$$R_P(f_{\hat{k}}) \le \min_{k \le K} R_P(f_k) + 2\sqrt{\frac{\log(2K/\alpha)}{2n_{\mathrm{val}}}}.$$

Choosing with validation consumes validation information. Final assessment uses the test sample.

The candidate list must be fixed relative to validation data for this proof.

Binary classification as a decision problem

Set $\mathcal{Y} = \mathcal{A} = \{0, 1\}$ and $\ell(a, y) = \mathbf{1}\{a \neq y\}$.

$$\eta(x) = \mathbb{P}(Y = 1 \mid X = x), \qquad R_P(h) = \mathbb{P}(h(X) \neq Y).$$

The conditional losses are

$$\rho(0 \mid x) = \eta(x), \qquad \rho(1 \mid x) = 1 - \eta(x).$$

The conditional distribution and the operational loss determine the action.

The Bayes classifier and its excess-risk identity

Choose ties in favor of class 1:

$$h_P^*(x) = \mathbf{1}\{\eta(x) \geq 1/2\}, \qquad R_P^* = \mathbb{E}\min\{\eta(X), 1-\eta(X)\}.$$

For every measurable classifier h,

$$R_P(h) - R_P^* = \mathbb{E}[|2\eta(X) - 1|\mathbf{1}\{h(X) \neq h_P^*(X)\}]\,.$$

Proof: condition on $X = x$ and subtract the smaller of the two conditional losses.

Errors near $\eta = 1/2$ carry small excess risk. Irreducible label noise remains in R_P^*.

Shared notation for Lectures 2 and 3

$Z = (X, Y),\ P$	A random example and its population distribution.
$S_n,\ n,\ m$	Training sample, training size, independent test size.
$\ell,\ R_P, \widehat{R}_n$	Loss, population risk, empirical risk.
α	Probability of failure of a theorem's event.
$f,\ h, \eta$	Predictor, binary classifier, conditional class probability.
$\mathcal{F}, \mathcal{H},\ v$	Predictor class, classifier class, VC dimension.
$p, \beta^*, \mathbf{X}$	Feature dimension, regression coefficient, fixed design matrix.
σ, λ	Noise standard deviation and regularization parameter.
c, C	Positive universal constants, allowed to change between bounds.

Appendix: the exponential-moment proof of Hoeffding

For independent $U_i \in [0,1]$, Hoeffding's lemma gives

$$\mathbb{E}e^{t(U_i - \mathbb{E}U_i)} \le e^{t^2/8}.$$

By Markov's inequality and independence,

$$\mathbb{P}\left\{\frac{1}{n}\sum_i (U_i - \mathbb{E}U_i) > u\right\} \le \inf_{t>0} \exp(-ntu + nt^2/8) = e^{-2nu^2}.$$

Apply the same argument to $-U_i$ and add the two tails.

The minimizer is $t = 4u$. Substituting $U_i = \ell(f(X_i), Y_i)$ requires f fixed relative to these observations.

Appendix C: conservative VC tail constants

Write $P_n g = n^{-1} \sum_i g(Z_i)$, $P'_n g = n^{-1} \sum_i g(Z'_i)$, and $Pg = \mathbb{E}_P g(Z)$. For $nt^2 \geq 2$, symmetrization gives

$$\mathbb{P}(\Delta_n > t) \leq 2\mathbb{P}\left\{ \sup_h |P_n g_h - P'_n g_h| > t/2 \right\}.$$

Conditional pair swaps produce $n^{-1} \sum_i \xi_i a_i$ with $|a_i| \leq 1$. Its two-sided tail at $t/2$ is at most $2e^{-nt^2/8}$. There are at most $\Pi_{\mathcal{H}}(2n)$ coefficient patterns. Hence

$$\mathbb{P}(\Delta_n > t) \leq 4\Pi_{\mathcal{H}}(2n) e^{-nt^2/8} \leq 8\Pi_{\mathcal{H}}(2n) e^{-nt^2/32}.$$

For $nt^2 < 2$, the last right-hand side exceeds 1, so the bound is trivial.

Appendix D: a sharper uniform bound

For a loss class $\mathcal{G}$ with $0 \le g \le 1$, symmetrization and bounded differences imply, with probability at least $1-\alpha$,

$$\sup_{g\in\mathcal{G}} |Pg - P_n g| \le 2\mathfrak{R}_n^{\mathrm{abs}}(\mathcal{G}) + \sqrt{\frac{\log(1/\alpha)}{2n}}.$$

For a VC loss class, a uniform covering bound and chaining yield

$$\log N(u, \mathcal{G}, L_2(Q)) \le Cv\log(A/u), \qquad \mathfrak{R}_n^{\mathrm{abs}} \le \frac{C}{\sqrt{n}} \int_0^1 \sqrt{v\log(A/u)}\, du.$$

The integral is finite, giving the order $\sqrt{v/n}$.

Q is any probability measure; N is the minimal $L_2(Q)$ covering number. Universal $A > 1$. Entropy/chaining are stated results.

Appendix E: testing and the classification lower bound

For distributions Q, Q' with densities q, q',

$$\mathrm{TV}(Q, Q') = \tfrac{1}{2} \int |q - q'|, \quad \mathrm{KL}(Q \| Q') = \int q \log(q/q'), \quad \mathrm{TV} \le \sqrt{\mathrm{KL}/2}.$$

Any test of Q versus Q' has equal-prior average error at least $(1 - \mathrm{TV})/2$.

Average over all $\omega \in \{-1, 1\}^v$ and pair opposite values of bit j, holding the other bits fixed. Each bit has error at least $1/4$. Sum over j:

$$\sup_{\omega} \mathbb{E}_{\omega}[\text{excess risk}] \ge \frac{1}{2^v} \sum_{\omega} \frac{\varepsilon}{v} \mathbb{E}_{\omega}[\text{wrong bits}] \ge \varepsilon/4.$$

Appendix: logistic loss and classification excess

For a probability predictor $q(x) \in (0,1)$, define

$$\ell_{\log}(q, y) = -y \log q - (1-y) \log(1-q).$$

Its excess population risk is

$$R_{\log}(q) - R_{\log}(\eta) = \mathbb{E}_X \, \mathsf{KL}\Big(\mathrm{Ber}(\eta(X)) \| \mathrm{Ber}(q(X))\Big).$$

The plug-in bound and Pinsker imply

$$\mathcal{E}_P(\mathbf{1}\{q \geq 1/2\}) \leq \sqrt{2\,[R_{\log}(q) - R_{\log}(\eta)]}.$$

Log loss is unbounded near 0 and 1. Clipping $q \in [a, 1-a]$ gives loss bounded by $\log(1/a)$, for $0 < a < 1/2$.

References for Lecture 2

- **[SL]** Shalev-Shwartz and Ben-David (2014), *Understanding Machine Learning*, Chapters 4, 6, 11, and 28. Author's book page
- **[EP]** Ryan Tibshirani, *Empirical process theory*, Berkeley Statistical Learning notes (2023). Course notes and sources
- **[RH]** Rigollet and Hütter, *High-Dimensional Statistics*, sections on testing and minimax lower bounds. Author notes, arXiv:2310.19244

Scope starts from the supplied integrated syllabus and follows the requested theoretical extension. Proofs use the notation and normalizations defined in these slides.